

Dataset-ingestion backlog — improving the recommender with external data

Produced **2026-06-19** via a multi-agent web-research sweep across the data categories that **Google Dataset Search** (<<https://datasetsearch.research.google.com/>>) surfaces — product/attribute, manufacturer/brand, standards/eligibility, compliance/trade, and pricing/commodity — cross-checked against our shipped ingestion pipeline. Every dataset below was found and vetted for **license, access method, and cost**, with **free sources prioritized** and **valuable paid sources flagged with their real price**.

Goal: feed better **attributes, datasheets, manufacturer hierarchy, cross-references, classification, compliance flags, and price signals** into the Meridian Product Finder so search, embeddings (#4), the data-quality score (#11), the cross-reference engine, the rebate estimator, and the landed-cost overlay all get sharper.

How to keep discovering datasets (the method)

Use **Google Dataset Search** with these query shapes, then vet each hit for license + access + freshness before adding a story here:

- `electrical product attributes`, `circuit breaker dataset`, `lighting fixtures dataset`
- `GTIN UPC product database`, `manufacturer brand ownership`, `company parent subsidiary`
- `ENERGY STAR certified products`, `DLC qualified products`, `UL listing`
- `HTS tariff section 301`, `REACH SVHC`, `RoHS substances`, `Prop 65 list`
- `producer price index electrical`, `copper aluminum commodity price`

Also worth bookmarking as ongoing discovery surfaces: **data.gov**,

data.europa.eu (EU Open Data), **AWS Registry of Open Data**, **Hugging Face Datasets**, **Kaggle**, and **github.com/awesomedata/awesome-public-datasets** /

awesome-industrial-datasets.

Two ingestion patterns (reuse, don't reinvent)

Every story picks one of the two patterns the app already ships:

A. Static build-time layer *(for stable reference/attribute data — the default)*

1. Land the source as JSON under `data/real/` (or a new `data/ref/<dataset>/`). 2. Extend / add a builder modeled on `scripts/build-real-products.mjs` (schema validate → link-verify URLs → dedupe → emit a typed `.ts` with a `verifiedAt`). 3. **Provenance + confidence gate** via `lib/catalog/provenance.ts` (`assessRecord` → only ≥95 `productionReady` enter the catalog; the rest land in the data-quality report). Tag each record's `dataSource` / `provenance` so it flows into the **#11 data-quality score**. 4. Merge in `lib/catalog/generate.ts` (a real SKU matching a curated entry *upgrades* it; otherwise it's appended) → it shows up in `getCatalog()`, search, embeddings, and faceting automatically. 5. Sync identifiers/hierarchy: keep brands aligned with `data/real/brand-hierarchy.ts`; populate `gtin` / `catalogNumber` / `wescoSku` / `parentCompany`; the quality report (`lib/catalog/data-quality.ts`) surfaces gaps.

B. Dormant live API seam *(for fresh/volatile data — pricing, availability, programs)*

Mirror the shipped seams (`lib/integration/*-live.ts`, e.g. `commodity-live`, `embeddings-live`): an `env()` gate + `xConfigured()` boolean, `{enabled:false}` fail-closed union, `rateLimit` → `requireApiAuth` → `dormant-gate` → `Zod` → `logic` → `logApiError`, a `/api/...` route, and a `health` flag. **\$0 and zero network until its key is set** — honors the cost guardrail.

License discipline: several sources (Octopart/Nexar, GS1, IDW, Icecat Full)

restrict caching/redistribution. Live/volatile data → pattern B (fetch per

request, don't persist). Openly-licensed reference data (ETIM ODC-By, ENERGY

STAR public-domain, Wikidata/GLEIF CC0) → pattern A (snapshot at build), keeping

the required attribution.

Ranked master list (value ÷ cost)

Free, highest-ROI (do first):

Dataset	License / cost	Access	What it improves
ETIM Classification Model	Free (ODC-By)	bulk CSV/IXF/Excel	The electrical attribute backbone — defines required specs per category; drives faceting, completeness scoring, like-for-like cross-ref
ENERGY STAR Certified Products	Free (public-domain)	Socrata REST + CSV	Cert flag + efficacy/wattage/UPC for lighting/electronics; spec + GTIN fill
DLC QPL (SSL/NLC)	Free (web export; bulk = paid)	CSV export	DLC-qualified flag + photometric specs → powers the shipped rebate estimator (#6)
GLEIF LEI (L1+L2)	Free (CC0)	bulk CSV/JSON + REST	Parent-company rollups + a stable global ID per manufacturer (brand hierarchy)
Wikidata (P127/P749, GTIN P3962)	Free (CC0)	SPARQL + dumps	Brand aliases + ownership chains + cross-walk IDs (LEI/CIK/DUNS) → entity resolution for cross-ref
Open Icecat	Free tier (Full = paid)	XML/JSON API + feeds	Datasheets/specs/images/GTINs/marketing copy on the real-products layer (strong on lighting/datacom)
USITC HTS + Section 301/232	Free (public-domain)	JSON/CSV (no key)	HTS code + duty + 301 surcharge → the shipped tariff/landed-cost overlay (#14)
California Prop 65 list (OEHHA)	Free	CSV/XLSX	Per-SKU Prop 65 flag (PVC/lead wire, brass connectors) + BOM rollup
ECHA REACH SVHC + EU RoHS	Free	CSV/web	RoHS/REACH compliance flags + BOM rollup
FCC Equipment Authorization (EAS)	Free (public-domain)	Socrata REST + bulk	"FCC ID verified" flag for wireless/datacom; grantee→manufacturer normalization
BLS PPI (electrical/metals series)	Free	REST API	Wholesale price-trend for YOUR products (wire/conduit/switchgear) — metal-cost passthrough
World Bank "Pink Sheet" + FRED	Free	CSV / REST	License-clean copper/aluminum benchmarks for the commodity strip (FRED already wired)
SEC EDGAR	Free (public-domain)	REST + bulk	US public-parent identity + former-name history (resolves rebrands/spin-offs)
Amazon ESCI Shopping Queries	Free (research)	GitHub/HF	Substitute/Complement ground-truth labels → train/validate the cross-ref + substitute engine and ranking
DSIRE (incentive programs)	Free open-data (commercial API = ask)	XML/JSON + API	The program layer — which rebates/incentives apply by location — pairs with QPL/ENERGY STAR product eligibility
NREL/OpenEI URDB (utility rates)	Free (dev key)	REST + CSV	Converts kWh-saved → \$ saved for the rebate estimator using the customer's local rate
DOE CCMS + Tax-Credit Lookup	Free (public-domain)	bulk Excel/CSV	Efficiency data for federally-regulated gear (transformers/motors) + IRA 25C tax-credit eligibility
UNSPSC	Free (PDF; member Excel)	PDF/data.gov	Procurement classification (already shipped #64) — keep current

Freemium (free tier; paid to scale — build dormant, pattern B):

Dataset	Free tier	Paid	Improves
Octopart / Nexar	1,000 calls/mo	quote-only (enterprise)	Live datacom/electronics pricing, datasheets, lifecycle/EOL, second-source
GS1 Verified by GS1	free single lookup	membership (~low-hundreds \$/yr+)	Authoritative GTIN/brand validation; barcode→true brand owner
TraceParts	free per-part	publishing plans	CAD/drawings + datasheet PDFs on product pages
metals.dev	100 req/mo	from ~\$1.49/mo	Near-real-time copper/aluminum tick for the strip
AHRI Directory	≤250 rows/search	data subscription	HVAC cert + rebate/tax-credit eligibility
UL Product iQ	free search	paid API	UL-listing verification (breakers/wire/conduit/connectors)

Paid watchlist (provision/quote before a story — pattern B, dormant):

Dataset	Price	Why it's worth it
IDEA Industry Data Warehouse (IDW)	quote-only (NEMA/NAED)	The domain-perfect source — the US electrical channel's own ETIM/UNSPSC-native attributes, datasheets, GTINs and price sync for breakers/wire/conduit. The single highest-value paid option.
Full Icecat	low-thousands \$/yr	Guaranteed coverage across 40k+ brands
OpenCorporates API	£2,250–£12,000/yr	Private/non-US manufacturer legal entities GLEIF/EDGAR miss
Crunchbase	\$99–199/mo (+Enterprise API)	Recent M&A/acquisition layer for brand hierarchy
D&B Direct+	5–6 figure/yr	Gold-standard complete corporate family tree
LME official data	~\$85/report, ~\$1,200/yr API	Official metal settlements (not justified vs free Pink Sheet for this app)

Sprint stories (value-ordered, all \$0 unless noted)

DI-S1 · Attribute + classification backbone *(free, pattern A)*

DI-1 — ETIM classification ingestion. Land ETIM (groups/classes/features/values/ units) as a typed reference under `data/ref/etim/`; map our subcategories → ETIM classes; derive **required-spec sets** per category. *Improves:* completeness scoring (#11), faceting, and like-for-like cross-ref matching. *Acceptance:* each electrical subcategory maps to an ETIM class; the data-quality score penalizes a breaker missing amperage/poles/breaking-capacity; ODC-By attribution shown.

DI-2 — ENERGY STAR + DLC QPL lighting enrichment. Snapshot ENERGY STAR (Socrata) + DLC QPL CSV; upgrade matching lighting SKUs with cert flag + efficacy/ wattage/UPC. *Improves:* rebate estimator (#6), compliance facet, spec fill.

Acceptance: lighting SKUs carry an `energyStar` / `dlcQualified` flag + photometric specs; rebate estimate cites the listing.

DI-3 — UNSPSC refresh. Re-parse the current UNSPSC codeset; verify `unspscFor` coverage. *Acceptance:* no electrical subcategory falls back to a generic code.

DI-S2 · Manufacturer / brand entity layer *(free CC0, pattern A)*

DI-4 — GLEIF LEI parent rollups. Ingest LEI L1+L2 RR; attach an LEI + direct/ ultimate parent to each manufacturer. *Improves:* brand hierarchy, second-source.

Acceptance: every catalog brand resolves to an LEI or is reported as a gap.

DI-5 — Wikidata brand graph. SPARQL for brand→owner (P127/P749) + aliases + cross-walk IDs; feed the brand-hierarchy registry + cross-ref "equivalent brand".

Acceptance: "Square D / Homeline / QO → Schneider" resolves; aliases improve search recall.

DI-6 — SEC EDGAR name history. Add US public-parent identity + former-name map.

Acceptance: a rebranded/spun-off manufacturer still resolves to its current parent.

DI-S3 · Compliance + trade enrichment *(free public-domain, pattern A)*

DI-7 — HTS + Section 301. Replace the static tariff table with the USITC HTS + Chapter-99 301/232 mapping; assign HTS per subcategory.

Improves: landed-cost overlay (#14) accuracy. *Acceptance:* duty + 301 surcharge derive from a real HTS code.

DI-8 — Prop 65 + REACH/RoHS flags. Ingest OEHA Prop 65 + ECHA SVHC + RoHS Annex II as CAS lists; flag SKUs whose materials reference a listed substance.

Acceptance: per-SKU compliance flags + BOM rollup ("contains SVHC?").

DI-9 — FCC EAS verified flag. Dormant Socrata seam (pattern B) OR build-snapshot for the grantee-code→manufacturer map. *Acceptance:* wireless/datacom SKUs show an "FCC ID verified" facet.

DI-S4 · Product content + price/demand signals *(free, mixed pattern)*

DI-10 — Open Icecat datasheets. Pattern A snapshot of Open Icecat for sponsored brands → fill datasheets/images/specs/GTINs/marketing copy on the real-products layer (respect DRM-gated assets). *Improves:* #11 score, embeddings (#4) chunks, search.

DI-11 — BLS PPI + Pink Sheet commodity signals. Add BLS PPI electrical series + World Bank Pink Sheet to the commodity strip (FRED seam already exists). *Improves:* metal-cost passthrough realism.

DI-12 — Amazon ESCI substitute labels. Use ESCI Substitute/Complement pairs as ground truth to **validate + tune** the cross-reference/substitute ranking (offline eval, not shipped data). *Acceptance:* a measured precision/recall on substitutes.

DI-13 — URDB + DSIRE rebate depth. Dormant seams (pattern B): URDB for \$-saved math, DSIRE for "which programs apply here". *Improves:* the rebate estimator from a range to a location-aware figure.

DI-S5 · Freemium dormant seams *(free tier, gate dormant, pattern B)*

DI-14 — Nexar/Octopart (datacom/electronics live pricing+datasheets — extends the shipped offer ladder; 1k free calls/mo, get a quote before production).

DI-15 — GS1 Verified GTIN validation (clean/verify GTINs; barcode→brand for scan-to-reorder + visual part ID).

DI-16 — TraceParts CAD + metals.dev live tick.

DI-S6 · Paid watchlist *(provision/quote first — biggest data lift)*

DI-17 — IDEA IDW (the domain-perfect attribute+price source for breakers/wire/ conduit — get a quote; dormant seam until funded). **DI-18 — Full Icecat,**

OpenCorporates, Crunchbase/D&B for deep entity coverage.

Notes & guardrails

- **Cost guardrail honored:** every freemium/paid source ships as a **dormant,

env-gated seam** (\$0 until a key is set). The free build-layer datasets add no runtime cost.

- **Provenance everywhere:** ingested records carry a `dataSource` /source URL so the

data-quality score (#11) reflects the upgrade and the quality report tracks coverage — better data → better embeddings (#4) → better recall.

- **License first:** never persist Octopart/Nexar/GS1/IDW pricing or catalog data

beyond what their ToS allows; those are pattern-B (per-request, not cached).

- Start with **DI-S1 + DI-S2** — ETIM + ENERGY STAR/DLC + GLEIF + Wikidata are the

highest value-per-dollar, all free/CC0, and they compound (better attributes and better entity resolution lift search, cross-ref, embeddings, and compliance at once).